AMD DEVELOPER HACKATHON: ACT III | TEAM: TEAM-9992 | 19 AUTOMATED TESTS

IncidentLens AMD

Evidence-Bound LLM Gateway Telemetry Triage with AMD TokenFactory Integration

The Core Problem

LLM gateways can suffer premature SSE termination, retry cascades, and unsafe cache writes. A model asked to triage those traces can also produce plausible explanations that are not tied to any supplied event.

The Evidence-Bound Fix

IncidentLens algorithmically parses gateway traces into an indexed evidence matrix (`[EVT-001]` , etc.), sends the bounded bundle to AMD TokenFactory when configured, and enforces a fail-closed citation gate on model responses.

AMD TokenFactory Adapter

When `AMD_API_KEY` is configured, the bounded evidence bundle is sent to the event-issued TokenFactory chat-completions endpoint. The public deployment stays in disclosed deterministic evaluation mode.

Built by **Siddharth Srivastava** (solo builder, GitHub: `sidsri14/incidentlens-amd`)

PRODUCTION VULNERABILITIES | AI GATEWAY FAILURE MODES

Gateway Failure Modes & The Evidence Gap

Why trace-grounded checks matter before an AI-generated incident report is shown

1. Premature SSE Disconnection

The fixture records a client abort before the completion marker. IncidentLens flags the incomplete stream so an operator can investigate cancellation and billing behavior without inventing downstream facts.

Abort at 490 ms

MEASURED IN THE INCLUDED TRACE FIXTURE

2. Runaway Retry Cascades & 429 Storms

Rapid retries without randomized exponential backoff can compound upstream provider rate limits. The included fixture records three HTTP 429 responses in a 225 ms window.

3 HTTP 429s in 225 ms

MEASURED IN THE INCLUDED RETRY FIXTURE

3. Unbound Model-Generated Triage

A generated incident report can contain plausible claims that are not tied to the supplied trace. IncidentLens rejects unknown event IDs and uncited claim units before showing the report.

Fail-Closed Gate Required

REJECT UNCITED OR UNKNOWN REFERENCES

SYSTEM ARCHITECTURE CITATION-BOUNDED PIPELINE

The 3-Pillar Evidence-Bound Architecture

Deterministic trace parsing, stable evidence indexing, and fail-closed citation checks

1 Deterministic Telemetry Parser

`src/trace_parser.mjs` parses raw SSE stream transitions, HTTP response codes, and socket lifecycles algorithmically:

- Flags premature disconnections before `[DONE]` sentinels.
- Detects retry storms with delta < 200ms without jitter.
- Identifies attempted cache writes on truncated streams.
- Calculates TTFT & latency SLA outlier thresholds.

2 Bounded Evidence Matrix

`src/evidence_matrix.mjs` converts parsed events into a stable, inspectable evidence table:

- Every trace event receives a canonical ID: `[EVT-001]`, `[EVT-002]`.
- Timestamps, durations, token counts, and error codes are bound.
- The bounded evidence bundle is dispatched to TokenFactory only when an API key is configured.

3 Fail-Closed Citation Gate

`src/citation_validator.mjs` inspects every sentence in the model's incident report before surfacing to SREs:

- Every root-cause claim must cite a valid `[EVT-xxx]` ID.
- Any uncited factual assertion is flagged and rejected.
- Any unknown event ID not in the evidence matrix causes immediate fail-closed denial.

AMD INTEGRATION AMD TOKENFACTORY

AMD TokenFactory Integration

Event-issued chat-completions adapter with explicit authenticated and deterministic modes

AMD TokenFactory Client (`src/amd_rocm_client.mjs`)

Direct connectivity to the event-issued TokenFactory inference endpoint:

- **Endpoint:** `https://developer.amd.com.cn/radeon/api/v1/chat/completions`
- **Event Team:** `team-9992`
- **Runtime:** Event-issued TokenFactory endpoint; provider metadata is recorded from the API response.
- **Default Model:** `DeepSeek-V4-Flash`, with a configurable model override.
- **Strict Evidence Prompting:** Enforces event ID citation requirements directly in system instructions.

Recorded Authenticated Test Run

19,614 ms
OBSERVED END-TO-END TOKENFACTORY LATENCY

809 Tokens
EVIDENCE SYNTHESIS & TRIAGE PLAN GENERATED

5 Event Citations
FIVE KNOWN EVENT REFERENCES ACCEPTED BY THE GATE

USER EXPERIENCE OPERATOR COCKPIT

Interactive Operator HUD & Failure Simulator

Interactive trace waterfall, anomaly visualization, and automated citation badges

🖥️ Operator Cockpit Features

- **Preset Scenario Simulators:** One-click premature-disconnect, retry-storm, and cache-inconsistency fixtures.
- **Telemetry Waterfall Table:** Inspects status codes, socket duration, TTFT, token counts, and anomaly badges.
- **Evidence Matrix Viewer:** Interactive display of indexed events with stable event IDs.
- **Citation Verification Badge:** Visual confirmation that claim units cite known IDs, or that the response was rejected.

🛡️ Security & Enterprise Hardening

- **Explicit Deployment Modes:** Local server binds to `127.0.0.1`; the public demo runs a separate no-key serverless evaluation path.
- **Content Security Policy:** Strict CSP headers, `X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`.
- **Bounded Input:** JSON-only requests are capped at 64 KiB and parsed into a constrained event schema.
- **Zero-Key Fallback:** Includes deterministic offline fixtures for instant evaluation without external credentials.

ENGINEERING RIGOR PUBLIC OSS PROTOTYPE

Verification, CI/CD & Open Source Delivery

Reproducible tests, multi-version CI, and a public MIT repository

✅ Automated Test Suite

19/19 passing tests using the native Node.js test runner (`node --test`):

- `test/trace_parser.test.mjs` : Premature disconnects, retry storms, cache hazards, latency outliers.
- `test/citation_validator.test.mjs` : Known-ID acceptance plus unknown and uncited claim rejection.
- `test/server.test.mjs` : Scenario dispatch and local HTTP analysis lifecycle.
- `test/vercel_handler.test.mjs` : No-key serverless mode and truthful mode labelling.

🚀 GitHub Actions CI Matrix

GitHub Actions Run #38794986491:

- **Linux (Ubuntu-latest):** Node 20 (Pass), Node 22 (Pass), Node 24 (Pass).
- **Windows (Windows-latest):** Node 20 (Pass), Node 22 (Pass), Node 24 (Pass).
- **Security Audit:** `npm audit` reports 0 vulnerabilities.
- **Public MIT Repo:** github.com/sidsri14/incidentlens-amd

Run locally: `npm test && npm start`